

Syllabus — INFO 521: Introduction to Machine Learning

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[Term / Year — no dates per design] · Modality/meeting pattern: [confirm — in-person / online-async / hybrid] Instructor: [posted per offering] · [email]

! Confirm before publishing (bracketed throughout)

Instructor of record + contact + office + office hours; TA (if any); term & modality; whether FCML / PML2 / IML remain as reference texts; and all UA policy wording — this syllabus *links* to the official UA templates rather than quoting them, so that language is never authored here. Assessment percentages reflect the approved redesign scheme; policy prose is adapted from the Fall 2025 INFO 521 syllabus (used with permission).

1 · Course Curriculum — what will we learn?

1.1 Course Description

Machine learning studies algorithms that adjust their internal parameters — that “learn” — to recognize patterns and make predictions from example data. This course introduces the foundations of machine learning and builds them into a single probabilistic arc: from fitting a model by minimizing **loss**, to estimating parameters by **likelihood**, to reasoning about them with a full **posterior**, to **approximating** inference when exact answers are out of reach, and finally to **discovering** structure without labels. Along the way students implement practical methods for regression, classification, clustering, and dimensionality reduction in Python, and learn to evaluate and communicate what those methods do — and where they fail — on real data.

1.2 Expected Learning Outcomes

Course Learning Outcomes. By the end of this course, students will be able to:

- **CLO1.** Identify and evaluate machine-learning approaches appropriate for regression, classification, clustering, and dimensionality-reduction problems.
- **CLO2.** Implement machine-learning algorithms in Python using industry-standard tools and workflows.
- **CLO3.** Compare and interpret model performance using appropriate evaluation metrics, validation techniques, and statistical reasoning.
- **CLO4.** Apply machine-learning methods to real-world datasets to develop predictive and descriptive models.
- **CLO5.** Communicate machine-learning results, limitations, and recommendations through written, visual, and technical presentations.

Course Objectives.

1. Introduce the core methods used in modern machine learning.
2. Give students experience implementing these methods as algorithms in Python.
3. Give students experience using and critically evaluating these methods on data.

1.3 Program-level relevance

Competencies. This course addresses the following College of Information Science competencies for the MS in Information Science and MS in Data Science programs:

- **MS1 – Computational Thinking:** students exercise decomposition, pattern recognition, abstraction, and algorithmic thinking.
- **MS2 – Data Collection, Analysis, and Interpretation:** students collect, manipulate, and analyze data at different scales and interpret the results properly.
- **MS3 – Communication of Analytical Results:** students communicate their findings to interdisciplinary audiences using appropriate visualizations and technical writing.

Other roles. This course can serve as a core course for the MSIS Machine Learning subplan; a general elective for the MSIS Human-Centered Computing subplan; an elective for the MSDS program and the PhD in Information; and part of a PhD Minor in Information.

1.4 Course Assessment Plan

Assessment in this course is **specifications-based** (“specs grading”): work is evaluated against clear pass/reviser criteria rather than partial-credit point-picking, and most components allow revision toward mastery. Two applied **projects** anchor the course; **homework units**, weekly **peer-engagement loops and discussions**, periodic **checkpoints**, and a **prerequisite quiz** support and assess the learning outcomes. See §3.1.3 for mechanics and its weights table; see the Course Schedule (§1.6) for when each lands.

1.5 Course Topics

The course is organized as seven modules along one arc — *loss* → *likelihood* → *posterior* → *approximation* → *discovery*:

1. **Foundations & Linear Regression** — the linear model, least squares, the normal equations.
2. **Learning Theory: Bias–Variance & Regularization** — model evaluation, cross-validation, the bias–variance tradeoff, ridge regularization.
3. **Probability, Likelihood & MLE** — probability review, the Gaussian, maximum likelihood, Fisher information and parameter uncertainty.
4. **Bayesian Inference** — priors, posteriors, conjugacy; the Beta–Binomial model; the conjugate Gaussian posterior; ridge as MAP.

5. **Approximate Inference & Bayesian Classification** — logistic regression and Newton’s method; the Laplace approximation; a taste of MCMC / Metropolis.
6. **Support Vector Machines & Clustering** — max-margin classification and kernels (concept level); K-means; Gaussian mixtures and EM.
7. **Dimensionality Reduction (PCA)** — eigen-decomposition of the covariance matrix, variance explained, projection, and the limits of linear methods.

(*Neural networks are the natural continuation of this material and are covered in the sequel, INFO 557; they are not part of INFO 521.*)

1.6 Course Schedule (week pacing — no fixed dates)

One module per week over the accelerated 7.5-week term, with the final half week reserved for the project close. A checkpoint quiz on a module’s derivations is due the week the module is taught and **gates the project milestone due the following week**, so the derivation always lands before the milestone that uses it. Module 3 has no interactive tool, so its written discussion carries that week’s loop on its own; Week 5 has no discussion because it is the project midpoint. Section numbers marked † are from the PML (2022) table of contents — verify against your printing.

Wk	Module & topic	Readings	Peer loop & discussion	Key work due
1	M1 · Linear Models & Least Squares	PML Ch.1 (§1.1–1.3, §1.5); MML Ch.2; MML Ch.9 (§9.1–9.2)	Least squares · Introductions	Prerequisite quiz; HW Units 0–1; The project assigned
2	M2 · Generalization	PML §4.7; PML §11.3†; MML Ch.5	Bias–variance & ridge · The tradeoff in your own words	HW Units 2a–2b; Checkpoint quiz 1.1 (normal equations); project brief read; teams formed
3	M3 · MLE & Parameter Uncertainty	PML Ch.2, Ch.3, §4.1–4.2 (incl. §4.2.5), §4.7; MML Ch.6	Discussion only · What likelihood buys us	HW Unit 3; Checkpoint quiz 1.2 (MLE); Milestone 1.1
4	M4 · Bayesian Inference	PML §4.6; Think Bayes Ch.2, Ch.4 (supp.: Ch.18); PML §11.7†	Bayesian updating · How the posterior contracts	HW Unit 4; Checkpoint quiz 1.3 (posterior); Milestone 1.2
5	M5 · Approximate Inference & Bayesian Classification	PML Ch.10 (§10.1–10.2; review Ch.9); PML §4.6.8; Think Bayes Ch.19; MML Ch.7 (optional)	Tool only (no discussion — project midpoint)	HW Unit 5; Checkpoint quiz 2.1 (Newton–Raphson); Milestones 1.3 & 1.4; Project midpoint: Part 1 due; Part 2 begins

Wk	Module & topic	Readings	Peer loop & discussion	Key work due
6	M6 · SVM & Clustering	PML Ch.21 (§21.3–21.4); MML Ch.11; PML Ch.17 (§17.3†, optional)	K-means · When clusters do and do not match labels	Checkpoint quiz 2.3 (k-means / PCA); Milestones 2.1 & 2.2 (Gate A)
7	M7 · PCA	MML Ch.4 (§4.2), Ch.10; PML Ch.20 (§20.1)	PCA · What each lens saw	Milestones 2.3 & 2.4 (Gate B)
7.5	Course close	—	—	The project due: milestone 2.5 and the Part 2 report

The Peer loop & discussion column names each week’s discussion; the prompts and the quality criteria are on the [Discussions](#) page. Readings are assigned before each module (~10–20 pages/module; see §2.5). Cross-validation (Wk 2) and SVMs (Wk 6) are treated at concept level in lecture. Reading links live on the [Schedule](#) page.

2 · Course Information — what do we need to get started?

2.1 Instructor information

- **Instructor of record:** [posted per offering] · College of Information Science · [email] · [office]
- **Teaching assistant:** [if any — name / email]

2.2 Course communications

- Announcements are posted on **D2L**; ask questions during class/office hours (§2.3) or by email.
- Put **[521]** in the subject line of course email so we can find it quickly. Email the instructor [and TA] together for the fastest response.

Response time. We typically reply within one business day; we generally do not reply on weekends or university holidays. You’ll be notified in advance of any scheduling issues that may affect response times.

2.3 Office Hours

Office hours are posted on the D2L course site under *Content / Course Admin*. [**Confirm times / format.**]

2.4 Course Prerequisites

- Calculus I–II (derivatives and integrals; MATH 129) and Linear Algebra (MATH 313).
- Programming experience (at least three courses; the course language is **Python 3**).
- Introduction to statistics and discrete/continuous probability (ISTA 311 recommended).

If you are missing a prerequisite (or took it long ago) you may still succeed, but expect to put in additional effort — please use office hours. An early **prerequisite quiz** serves as a self-assessment for you and a calibration for the instructor. Multivariate calculus is helpful but not required; relevant concepts are introduced as needed.

2.5 Course Format and Session Structure

- **Modality:** [confirm — in-person / online-async / hybrid].

- **Weekly rhythm:** each module delivers its lecture material (slides/notes, with interactive in-lecture tools), an assigned reading, a **peer-engagement loop**, and a **homework unit**.
- **Peer-engagement loops** run in four stages across the week: (1) work through the module’s interactive tool and pass its built-in mastery check (the loop’s knowledge check); (2) post your own answer to that week’s published discussion prompt in your team’s D2L discussion; (3) give structured peer review to **every member of your team**, one response each; (4) respond to the reviews on your own post and revise. **Credit requires completing all four stages, including replying to all of your teammates.**
- **Discussion quality.** Posts and responses are held to a published standard, not just a word count. Your post must make a claim, support it with a concrete result from your own work that week, state the reasoning that connects them, and name one limit. Each response must engage a specific claim your teammate made, then extend it, challenge it with a stated reason, or connect it to other work. Agreement without addition does not earn credit. The six prompts and the full criteria are on the [Discussions](#) page.
- **Teams.** You stay in the same project team for the whole term, so the teammates who review your Week 2 post are the people you build both projects with.
- **Readings:** $\approx 10\text{--}20$ pages before each module. Estimate ≈ 1 hour per 10 pages (<https://cte.rice.edu/resources/workload-estimator>). Tell us if that estimate is off.
- **Content:** lecture materials, external links, and textbook excerpts are posted on D2L; the interactive activities are hosted online and also embedded in D2L.

2.5.1 Workload. Plan on a **minimum of 5–6 hours per week** beyond scheduled lecture time. Students spending substantially more should come to office hours; feedback on workload is welcome.

2.6 Required Resources and Equipment

2.6.1 Required texts. All required texts are available at no cost. We refer to them by the acronyms below in the schedule and readings.

Acronym	Title	Author(s)
PML	Probabilistic Machine Learning: An Introduction (2022)	Kevin Murphy
MML	Mathematics for Machine Learning	Deisenroth, Faisal & Ong
Think Bayes	Think Bayes (2nd ed.)	Allen B. Downey
UDL	Understanding Deep Learning	Simon J. D. Prince

Notation follows PML throughout the course. **[Confirm whether to retain FCML, PML2, and IML as optional reference texts, as in prior offerings.]**

2.6.2 Hardware and software. You will need a computer with:

- **Python 3** (3.11 recommended) and pip — <https://www.python.org>
- **Git** and a **GitHub account** (assignments are distributed via Classroom 50) — <https://git-scm.com>
- A **TeX distribution** (or Overleaf) for compiling PDFs — <https://www.ctan.org/starter>
- Access to **D2L** for announcements, discussions, and embedded activities.

No laptop? Borrow one from the library: <https://lib.arizona.edu/borrow/tech>.

2.7 Student Success Resources

- **UA resources** (accessibility & accommodations, campus safety, Campus Pantry, Campus Health, CAPS, Dean of Students Student Assistance, Survivor Advocacy): <https://catalog.arizona.edu/syllabus-policies>

- **Think Tank** — math, writing, and general tutoring: <https://thinktank.arizona.edu>
- **iSchool peer tutoring** — for course support, ask in the tutoring center’s Slack #general channel whether a tutor covers this course.

3 · Course Policies — what can we agree on to learn together?

3.1 Progress and Completion

3.1.1 Participation. Participation in this course means engaging in the weekly peer-engagement loops in good faith: completing the module activity, answering the week’s discussion prompt on time, giving constructive, respectful peer review to **every member of your team**, and responding to the reviews you receive. A loop earns credit only when all stages — including replies to all teammates — are complete. See the climate guidelines in §3.2.2.

3.1.2 Late work, revision, and missed assessments. This course is built around revision, so the policies differ from a traditional exam course:

- **Homework** and **project** work is specs-graded: submissions receive **Pass** or **Not-Yet**, and Not-Yet work may be **revised and resubmitted** under the mechanics in §3.1.3. Automated “correctness” checks may be resubmitted without limit; human-reviewed components use a limited pool of revision tokens.
- **Peer-loop** stages are time-bound within the week because they depend on teammates; missing a stage cannot be made up after the loop closes. The category (10%) is weighted to absorb an occasional miss rather than to punish one.
- **Checkpoints** are short open-book D2L quizzes on the module’s derivations. The auto-graded portion resolves on submission; written responses allow one revision. Each gates the project milestone due the following week; see §3.1.3.
- Documented medical and family emergencies are always accommodated — contact the instructor.

3.1.3 Grading.

3.1.3.1 Categories and weights.

Type	Category	Description	%
Summative	Project	One project in two parts on one dataset: Part 1 <i>One Problem, Three Lenses</i> (due Week 5, the midpoint) and Part 2 <i>From Inference to Discovery</i> (due Week 7.5), nine milestones gated by checkpoint quizzes.	55
Formative	Homework	Unit 0 (required) plus six core units of math + programming exercises, specs-graded with revision. Best five core units count (one drop built in).	25

Type	Category	Description	%
Summative	Checkpoints (×5)	Short open-book D2L quizzes (about 20 to 30 minutes each) on the module's derivations. One due per week, Weeks 2–6, each covering the module just taught, and each gating the project milestone due the following week . Mostly auto-graded, with two or three short written responses graded Satisfactory / Not-Yet with one revision. The coding verification that once sat beside these derivations now lives in the homework units and milestones, so nothing is assessed twice.	9
Formative	Peer-engagement loops & discussions	Weekly tool + answer to the week's discussion prompt + a response to every teammate + respond & revise. Held to a published quality standard, not a word count.	10
Formative	Prerequisite quiz	Early self-assessment on course prerequisites.	1

3.1.3.2 Specifications grading. Each assignment lists explicit specifications; work that meets them earns a **Pass**, work that doesn't earns **Not-Yet** with feedback and a path to revise. Within each component, the letter is earned by the **bundle** of work completed at the Pass level (an A–E ladder defined per component in the assignment's grading guide), not by accumulating partial credit. The components then combine by the weights in §3.1.3.1. Autograded correctness may be resubmitted freely; human-reviewed revisions draw on a limited token pool, and a small independent-miss allowance keeps a single slip from cascading. Full mechanics are in each assignment's grading guide and the course GRADING reference.

3.1.3.3 Grading scale.

Grade	Range
A	90% ≤ cumulative

Grade	Range
B	80% ≤ cumulative < 90%
C	70% ≤ cumulative < 80%
D	60% ≤ cumulative < 70%
E	cumulative < 60%

3.1.3.4 No “pre-grading.” The instructor will not review an entire assignment before the due date, but is glad to answer **specific** questions. For code, isolate the problem to a line or behavior first (use print statements!) and be ready to explain what you’ve already tried.

3.1.3.5 Grade appeals. To contest a score, email a written explanation within one week of receiving feedback. After final letter grades post, follow the UA grade-appeal procedure: <https://catalog.arizona.edu/policy/courses-credit/grading/grade-appeal>.

3.1.3.6 Rounding. Cumulative scores within 1% of a cutoff are rounded up (e.g., 89.5% → A) for students who have completed all assignments and loops, barring documented emergencies.

3.1.3.7 Scaling. If assessments prove harder than intended, grades may be scaled — only ever **upward**, never down. Scaling is not guaranteed (and unlikely unless the class average is below a C), so proceed as if there will be none.

3.2 Classroom Climate and Community Engagement

3.2.1 UA policies and student resources. All UA courses adhere to the general UA policies: <https://catalog.arizona.edu/policies>. Links to the Threatening Behavior policy, Nondiscrimination and Anti-Harassment policy, Safety on Campus, and the Subject-to-Change statement are here: <https://catalog.arizona.edu/syllabus-policies>. [Use the current official UA syllabus-policy language/links for the offering term.]

3.2.2 Course climate and inclusion. Students and instructors share responsibility for a safe, welcoming, inclusive environment where everyone can challenge themselves to succeed. Because much of this course runs through team peer review, be especially considerate in written feedback: critique the work, assume good faith, and give teammates specific, actionable, respectful comments. Disruptive or disrespectful conduct may be referred to the Dean of Students.

3.2.3 Academic integrity. The UA Code of Academic Integrity (<https://deanofstudents.arizona.edu/policies/code-academic-integrity>) and the College of Information Science Academic Integrity Policy (<https://infosci.arizona.edu/students-and-career/academic-integrity>) apply to this course.

Not allowed:

- Using course-sharing sites (Chegg, Course Hero, OneClass, StuDocu, Numerade, etc.).
- Copying text (one sentence or more) verbatim or inadequately paraphrased from any source; see <https://lib.arizona.edu/research/write-cite/plagiarism>.
- Copying solutions to homework or projects from online sources or from students who took the course previously (including solutions posted to GitHub).
- Having someone else do any part of your work.
- Including figures you did not create yourself.
- Modifying the provided automated tests to make an incorrect solution pass, or fabricating, sharing, or reusing another student’s peer-loop completion token.

Allowed:

- **Generative AI tools**, provided you **disclose** how you used them (you will not lose points for disclosing). LLMs can hallucinate and reflect biases — you are responsible for verifying and critically examining anything you submit. Do not enter confidential information into prompts.
- **Collaboration** on homework, provided you name your collaborators in the submission and write up the work **in your own words**. Genuine peer review of teammates' loop reflections is expected and encouraged.
- Getting writing help from the UA Writing Center: <https://thinktank.arizona.edu/writing-center>.

Citing sources. Cite every resource you use — websites (with URLs), books, articles, and any human other than the instructors. Citing “Google” is not sufficient; cite the actual resource.

Source of truth. This syllabus is the canonical source for the course's academic-integrity policies; if a verbal statement conflicts with it, the syllabus governs. If a policy is unclear or has a loophole, tell the instructor; fixes are not applied retroactively.

Sanctions. Plagiarism is reported to the Dean of Students with sanctions commensurate with severity; the typical sanction is a failing grade in the course.

Policy and resource language adapted, with permission, from the Fall 2025 INFO 521 syllabus. Assessment structure, learning outcomes, competencies, topics, and schedule reflect the current course redesign.